V. Eswar Gupta
EE23B085

EE2019 Analog Systems – Quiz 3

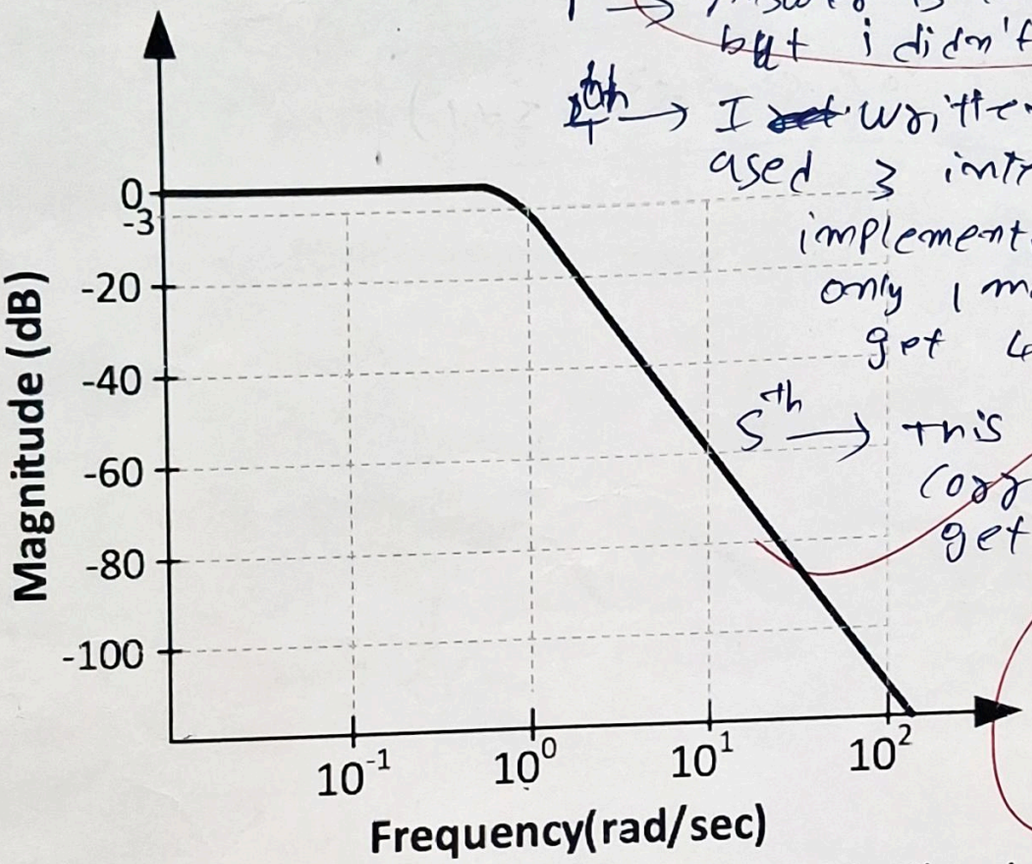
(Thursday, April 24th, 2025 – 6:15 to 7:30pm) given

Maximum Marks = 15; Time = 75 minutes

1st → Answer is 100% correct but i didn't got marks

4th → I ~~did~~ written correct only used 3 integrators & implemented but I got only 1 mark i need to get 4 marks

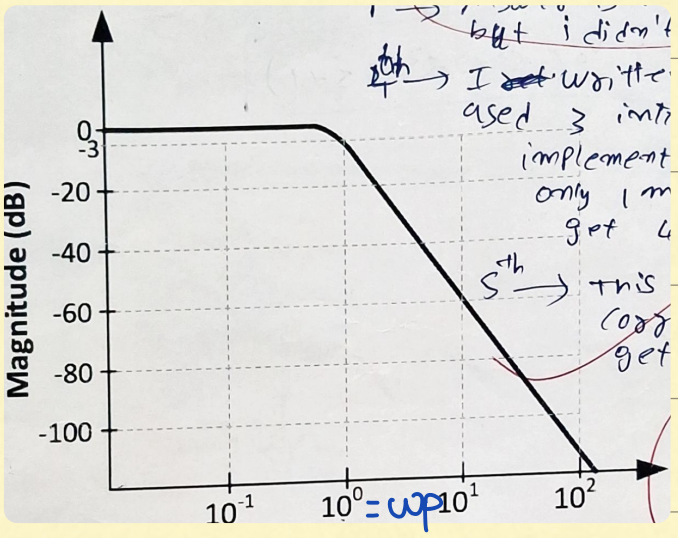
5th → this is also 100% correct but I didn't get marks



12/15

The magnitude response of a Butterworth Filter is shown.

1) Determine the transfer function of the filter and its magnitude **[2 marks]**
2) Show the location of poles on the s-plane with values marked on the real and imaginary axes **[2 marks]**
3) Plot the phase response of the filter and carefully mark important values on the x and y axes **[3 marks]**
4) Design a Butterworth Filter for the transfer function obtained in (1) using opamp-RC implementation **[4 marks]**
5) Design a Butterworth Filter for the transfer function obtained in (1) using Gm-C filter implementation **[4 marks]**



a) $|H(j\omega)|^2 = \frac{1}{1+(\omega/\omega_p)^6}$

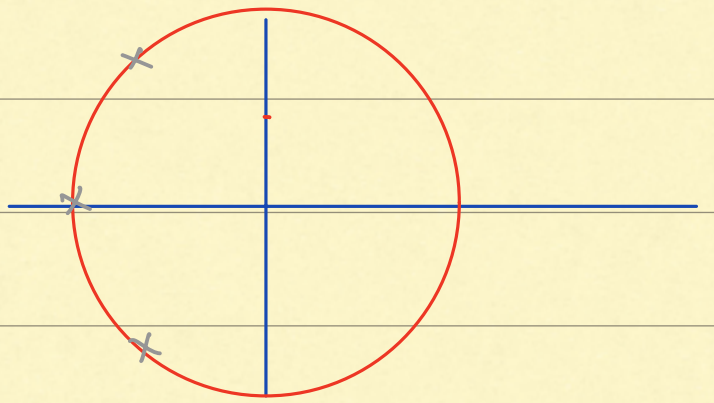
$|H(j\omega)| = \frac{1}{(1+\omega^6)^{1/2}}$

$|H(j\omega)|^2 = H(s)H(-s) = \frac{1}{1-s^6}$

$s = -1, -\left(\frac{1}{2}+\frac{\sqrt{3}j}{2}\right), -\left(\frac{1}{2}-\frac{\sqrt{3}j}{2}\right)$

$H(s) = \frac{1}{(1+s)(s+1/2+\sqrt{3}j/2)(s+1/2-\sqrt{3}j/2)} = \frac{1}{s^3+2s^2+2s+1}$

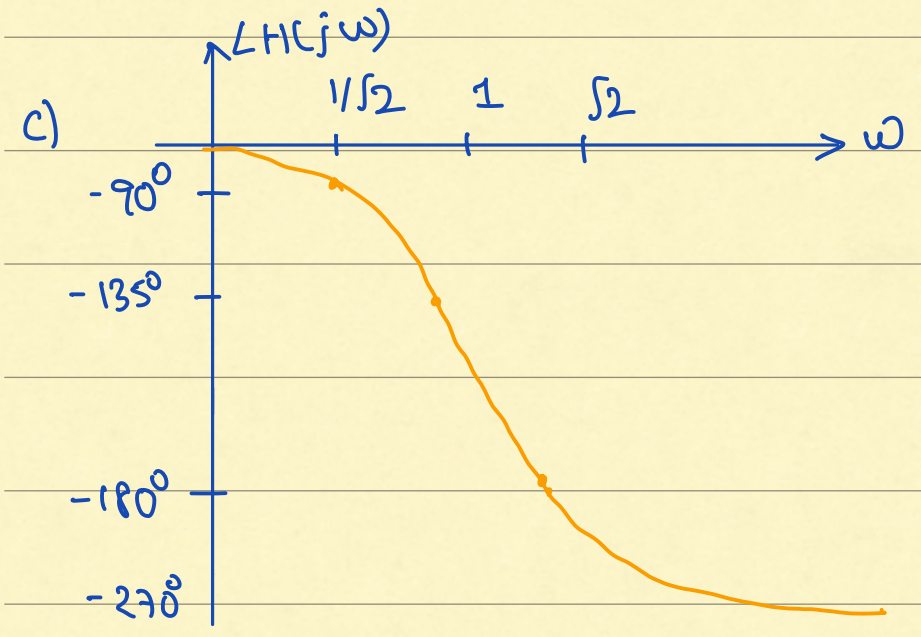
b)



$\frac{1}{1-2\omega^2+j(2\omega-\omega^3)}$

$\angle H(j\omega) = -\tan^{-1}\frac{2\omega-\omega^3}{1-2\omega^2}$

c)



d) 4) Design a Butterworth Filter for the transfer function obtained in (1) using opamp-RC implementation **[4 marks]**

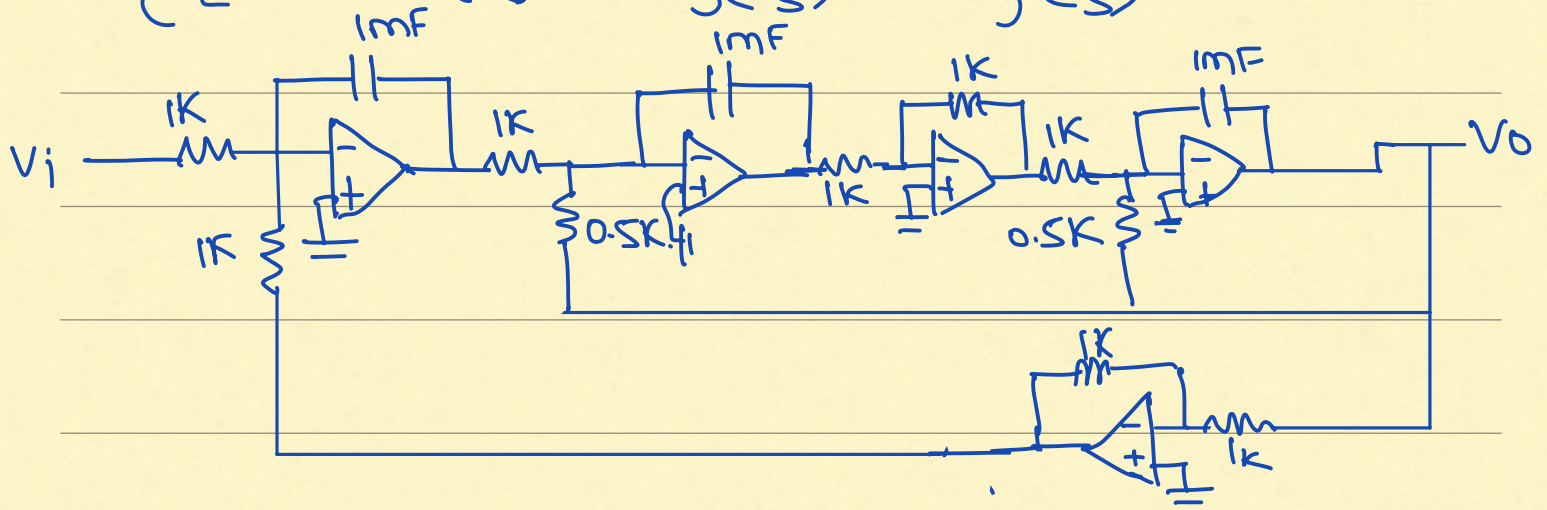
$$\frac{V_o}{V_i} = \frac{1}{s^3+2s^2+2s+1}$$

$$V_o(s^3+2s^2+2s+1) = V_i$$

$$(V_i - V_o)\left(\frac{1}{s}\right) = V_o(s^2+2s+2)$$

$$\left[(V_i - V_o)(-1/s) + 2V_o\right](-1/s) = V_o(s+2)$$

$$\left\{-\left[(V_i - V_o)(-1/s) + 2V_o\right]\left(\frac{-1}{s}\right) + 2V_o\right\}\left(\frac{-1}{s}\right) = V_o$$



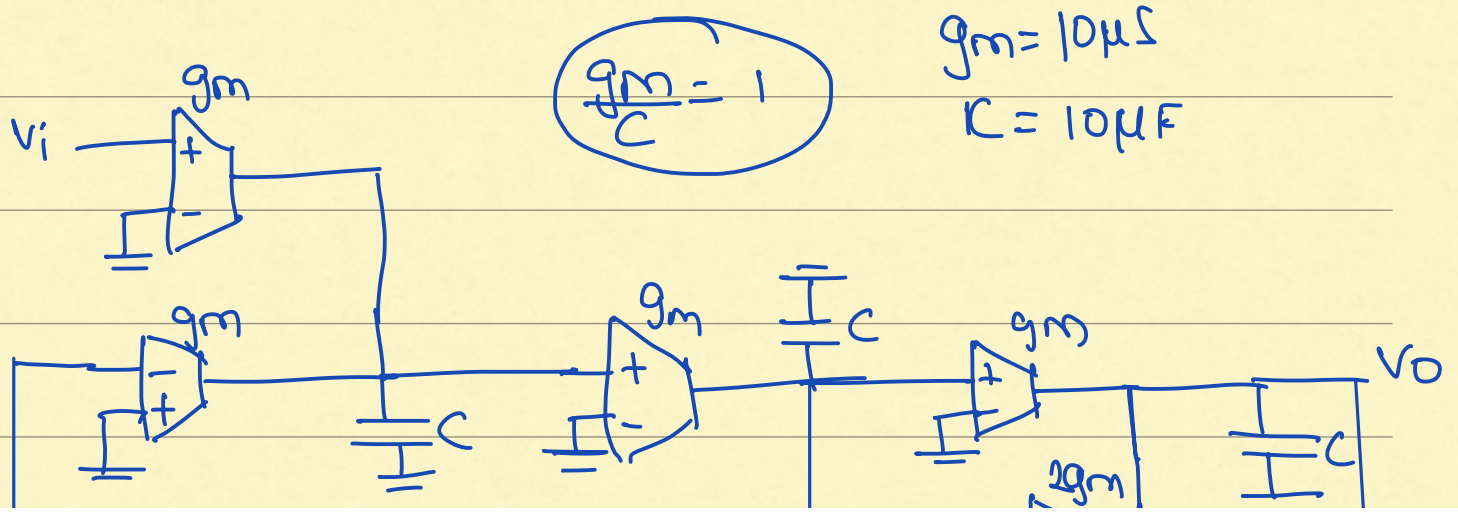
5) Design a Butterworth Filter for the transfer function obtained in (1) using Gm-C filter implementation **[4 marks]**

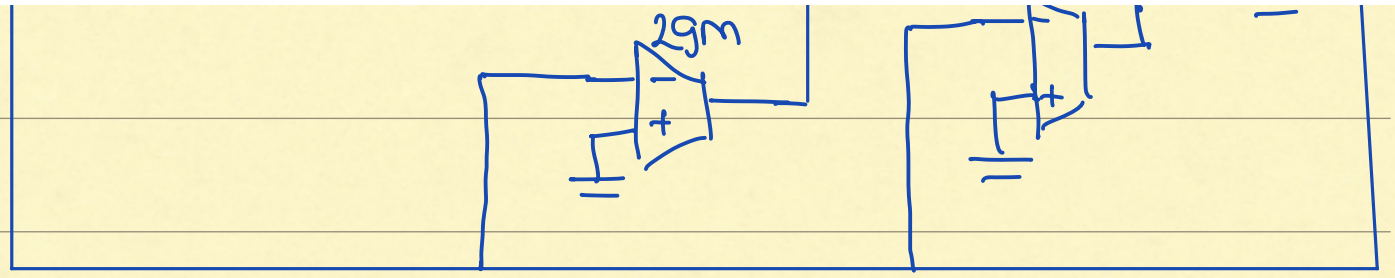
$$\left\{\left[(V_i - V_o)\,1/s - 2V_o\right](1/s) - 2V_o\right\}(1/s) = V_o$$

$\frac{g_m}{C} = 1$

$g_m = 10\mu S$

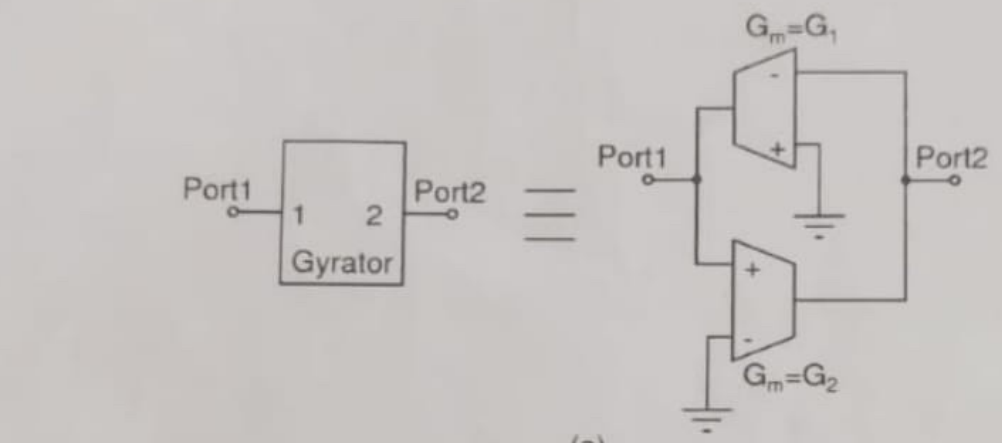
$C = 10\mu F$



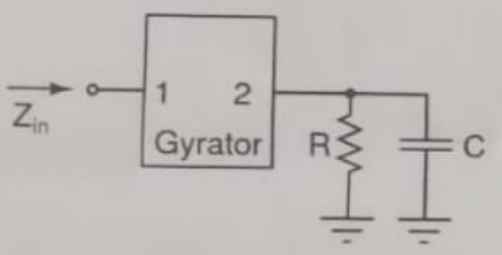

2gm

EE2019 Analog Systems – Quiz 4

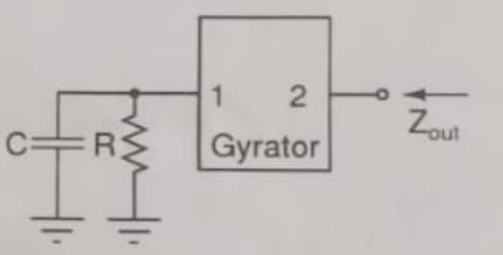
(Thursday, April 13th, 2023 – 8 to 8:50am)
Maximum Marks = 10; Time = 50 minutes



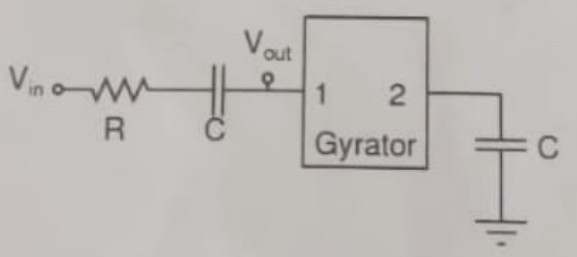
(a)



(b)



(c)

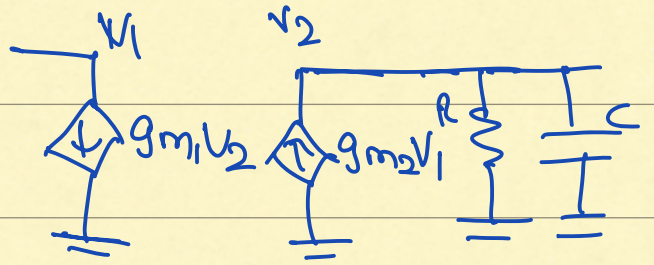


(d)

Figure (a) above shows the internal circuit of an asymmetric Gyrator. Figures (b), (c) and (d) use the gyrator from (a) in various configurations. G1 = 1mS; G2 = 2mS; R = 1kΩ; and C = 1nF.

1) Determine the input impedance Z_{in} of circuit shown in (b). **[2 marks]**
2) Determine the output impedance Z_{out} of circuit shown in (c). **[2 marks]**
3) Determine $V_{out}(s)/V_{in}(s)$ for the circuit shown in (d). **[3 marks]**
4) Determine the –3dB bandwidth, ω_n and Q of $V_{out}(s)/V_{in}(s)$ of figure (d). **[3 marks]**

a)



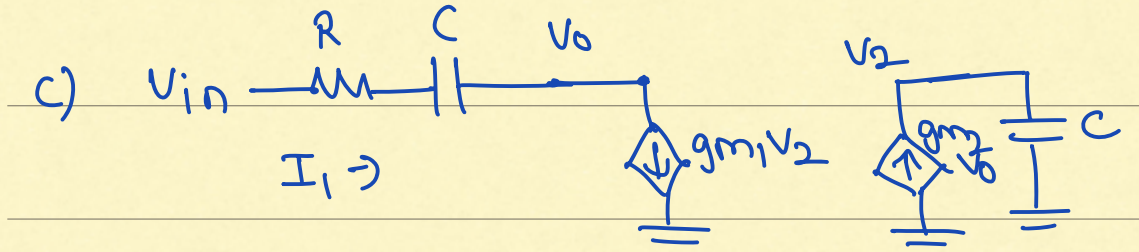
$$g_{m2}V_1 = V_2/R + V_2 SC$$

$$\Rightarrow V_1 = \frac{V_2(1+SRC)}{g_{m2}R}$$

$$Z_{in} = \frac{1+SCR}{g_{m1}g_{m2}R} = \frac{1+5\times10^{-6}}{2\times10^{-3}} = 500+5\times10^{-4}S$$

b) same

c)



$$V_2 = \frac{g_{m2}V_o}{SC}$$

$$I_1 = \frac{g_{m1}g_{m2}V_o}{SC} \qquad V_{in} - V_o = \frac{g_{m1}g_{m2}V_o}{SC}\left(R+\frac{1}{SC}\right)$$

$$V_{in} = V_o\left[1+\frac{2\times10^3}{S}\left(10^3+\frac{10^9}{S}\right)\right]$$

$$\frac{V_o}{V_i} = \frac{S^2}{S^2+2\times10^6 S+2\times10^{12}}$$

d

$$\omega_n/Q = 2\times10^6 \qquad \omega_n^2 = 2\times10^{12} \qquad \omega_n = 1.414\times10^6 = \omega_{3dB}$$

$$Q = 1/\sqrt{2}$$

$$\frac{(V_i - V_o)}{R + SL} = V_o SC + \frac{V_o}{R}$$

$$V_i - V_o = V_o SCR + V_o S^2 LC + V_o + V_o SL/R$$

$$V_i = V_o\left(S^2 LC + \frac{SL}{R} + SCR + 2\right)$$